TMDB MOVIES DATA ANALYSIS

Exploratory Data Analysis Report

Full Dataset Analysis | 1,375,553 Movies

Source: TMDB Movies Dataset 2023 (Kaggle)

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Report Summary

Dataset: 1,376,742 raw records
After cleaning: 1,375,553

Analysis Sections:
Univariate | Bivariate | Categorical | Statistical | Time Series

Language & Libraries:
Python (Pandas, Numpy, Matplotlib, Seaborn, Plotly)

Key Topics :

- Dataset Overview & Methodology
- Data Cleaning & Preprocessing
- Univariate Analysis
- Categorical Feature Analysis
- Bivariate & Multivariate Analysis
- Statistical & Correlation Analysis
- Runtime Analysis
- Time Series Analysis
- Key Findings & Conclusions

1. Dataset Overview & Methodology

This report presents a comprehensive Exploratory Data Analysis (EDA) of the TMDB (The Movie Database) Movies Dataset. The analysis was performed on the complete dataset of 1,376,742 movie records, covering films from 1865 to 2058. The objective is to uncover patterns in budget, revenue, genre popularity, production geography, and temporal trends in the global film industry.

1.1 Dataset Specifications

Attribute	Value
Source	Kaggle – TMDB Movies Dataset 2023 (930K+ Movies)
Raw Records	1,376,742
Total Columns	20
Post-Deduplication Records	1,375,553
Duplicate Records Removed	1,189
Date Range	1865 – 2058
Primary Language	English (largest group)

1.2 Column Schema

Column	Type	Description
id	int64	Unique TMDB movie identifier
title	object	Movie title
vote_average	float64	Average user rating (0–10)
vote_count	int64	Number of user votes
status	object	Production status (Released, In Production, etc.)
release_date	datetime64	Date of theatrical release
revenue	int64	Box office revenue (USD) – 0 = missing
runtime	int64	Film runtime in minutes – 0 = missing
budget	int64	Production budget (USD) – 0 = missing
original_language	category	Original language code (ISO 639-1)
genres	object	Pipe-separated genre list
production_companies	object	Production company names
production_countries	object	Countries of production
keywords	object	Associated thematic keywords
popularity	float64	TMDB proprietary popularity score
overview	object	Movie synopsis
profit	float64	Derived: revenue – budget

Column	Type	Description
roi	float64	Derived: (profit / budget) × 100
is_profitable	bool	Derived: profit > 0
primary_genre	object	First genre from genres list

2. Data Cleaning & Preprocessing

The raw dataset required several preprocessing steps to ensure analytical validity. Key transformations included duplicate removal, zero-to-NaN replacement for financial fields, data type conversions, and feature engineering.

2.1 Missing Value Analysis

The table below summarizes missing values across all columns after initial loading. High missingness in homepage and tagline columns led to their removal.

Column	Missing Values	Percentage (%)
homepage	1,234,668	89.68%
tagline	1,185,032	86.08%
imdb_id	718,129	52.16%
primary_genre	598,063	43.44%
overview	312,375	22.69%
release_date	294,354	21.38%
title	18	0.001%
keywords	0	0.00%
genres	0	0.00%
popularity	0	0.00%

2.2 Zero-to-NaN Replacement (Financial & Metric Fields)

Zero values in financial and metric columns were replaced with NaN, as 0 in these fields represents missing data rather than a true zero value.

Column	Zeros Replaced → NaN	% of Total Records
budget	1,297,858	94.27%
revenue	1,351,879	98.19%
runtime	419,438	30.47%
vote_average	1,018,260	73.96%

2.3 Cleaning Pipeline Summary

Preprocessing Steps Performed
1. Removed 1,189 duplicate records (by id field) → Final: 1,375,553 unique movies
2. Dropped homepage and tagline columns (>86% missing, low analytical value)
3. Replaced 0 → NaN in budget, revenue, runtime, vote_average 4. Converted release_date from object → datetime64[ns] 5. Converted status and original_language to category dtype (memory efficiency) 6. Filled keywords NaN with empty list [] for NLP safety 7. Filled production_companies, production_countries, spoken_languages with "Unknown"
8. Filled overview NaN with "No Overview Available" 9. Engineered features: profit, roi, is_profitable, release_year, release_month, primary_genre

2.4 Popularity Normalization

The raw popularity scores had extreme range (Min: 0.0, Max: 2,994.357). Min-Max Scaling was applied to normalize values to the [1, 10] range, enabling meaningful comparison with vote_average.

Metric	Before Scaling	After Scaling
Minimum	0.0	1.0
Maximum	2,994.357	10.0
Scale Range	Raw TMDB Score	Normalized [1–10]

3. Univariate Analysis

Univariate analysis examines the distribution of individual variables. This section covers the distribution of key numeric features: budget, revenue, and vote average.

3.1 Distribution of Budget

The budget distribution is highly right-skewed due to a small number of extremely high-budget blockbusters. A log transformation reveals a more interpretable, approximately normal distribution centered around mid-range production budgets.

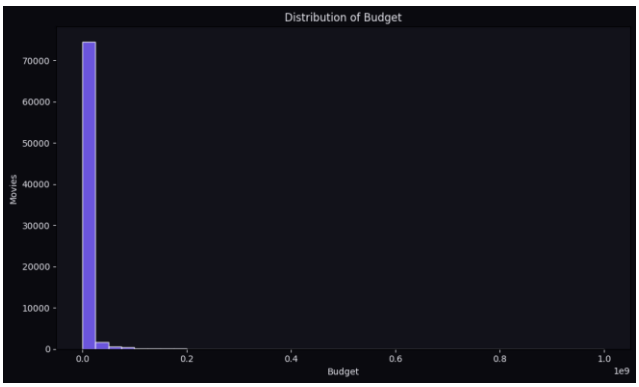


Figure 3.1a – Budget Distribution (Raw Scale)

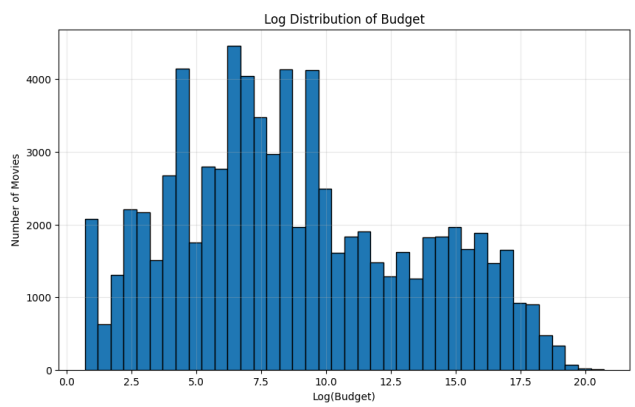


Figure 3.1b – Budget Distribution (Log Scale)

Key Insights – Budget Distribution
The raw budget distribution is heavily right-skewed due to a few mega-budget productions.
Most movies are produced on relatively modest budgets; the majority fall well below $100M.
Log transformation reveals a near-normal distribution, indicating a log-normal budget structure.
Extreme outliers at the high end inflate the mean, making median a better central tendency measure.

3.2 Distribution of Revenue

Revenue shows a similar right-skewed pattern to budget. The log-transformed distribution is bell-shaped, indicating log-normal revenue distribution consistent with many economic variables.

Figure 3.2a – Revenue Distribution (Raw Scale)

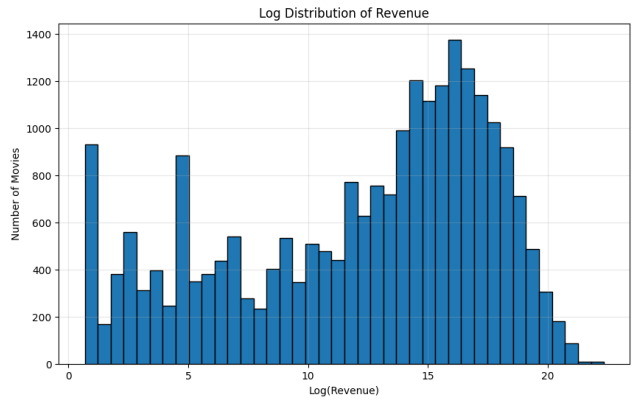
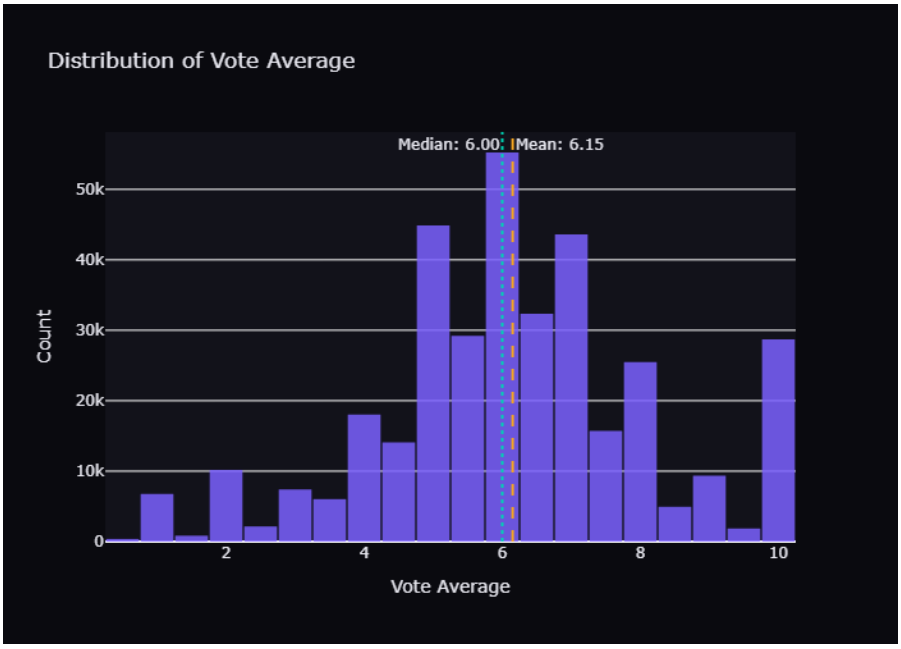


Figure 3.2b – Revenue Distribution (Log Scale)

Key Insights – Revenue Distribution
Revenue is even more skewed than budget; blockbuster films generate disproportionate earnings. Log scale transformation confirms a log-normal distribution pattern.
98.19% of revenue values were 0 in the raw data, reflecting missing financial disclosure. Only ~1.8% of films in the dataset have reported revenue figures.

3.3 Distribution of Vote Average

The vote average distribution reveals audience rating patterns across the full dataset. The distribution is left-skewed (right-heavy), indicating that most movies receive above-average ratings when voted on.



Statistic	Value
Mean Vote Average	6.150
Median Vote Average	6.000
Distribution Shape	Left-skewed (right-heavy tail)
Most Common Range	6.0 – 7.0
Interpretation	Most rated movies receive above-average scores

Key Insights – Vote Average
Mean: 6.15 and Median: 6.0 are very close, suggesting a moderately symmetric distribution around the center. The distribution is right-skewed (left tail), meaning most movies cluster around 6-8.
Very few movies receive extreme scores (below 2 or above 9), indicating rating regression to the mean. This reflects TMDB's audience which skews toward engaged, interested viewers who rate more generously.

4. Categorical Feature Analysis

This section explores the distribution of categorical variables including original language, movie genres, production companies, and production countries.

4.1 Top 15 Original Languages

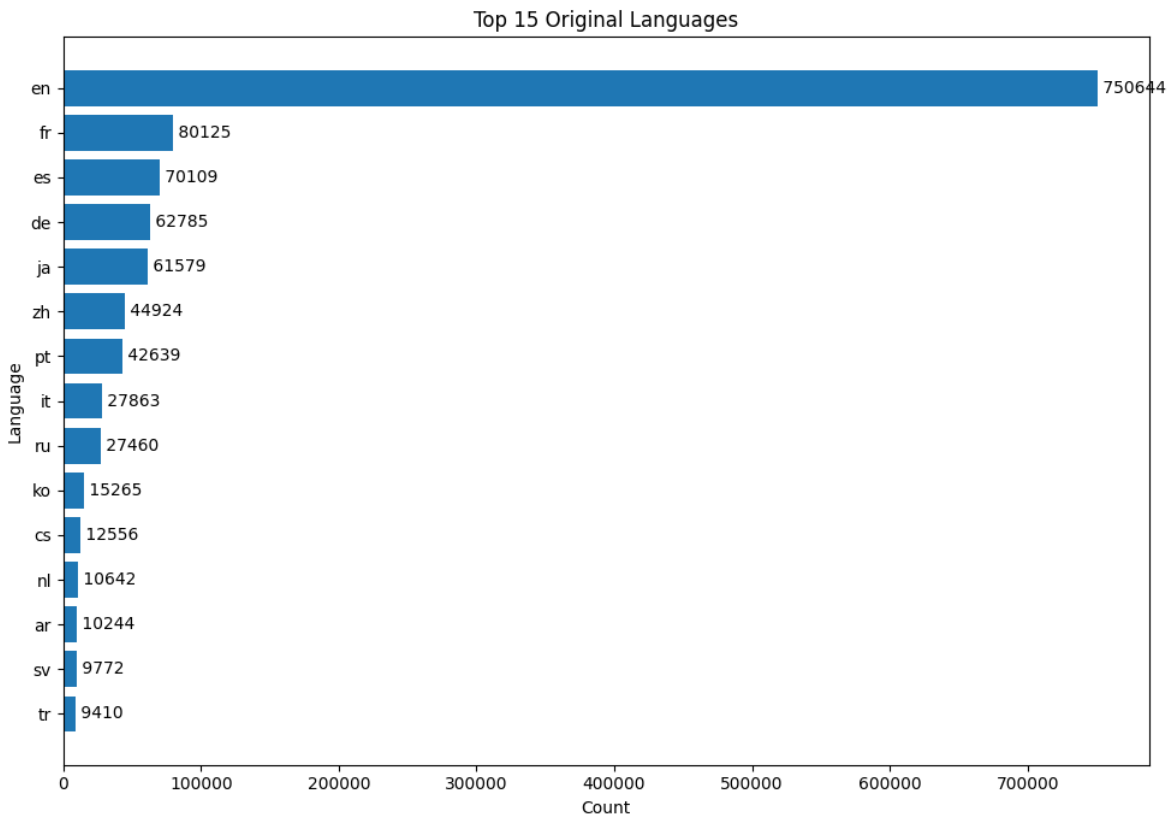


Figure 4.1 – Top 15 Original Languages by Movie Count

Key Insights – Language Distribution
English dominates the dataset, reflecting Hollywood's global production output and TMDB's US-centric user base. Japanese, French, Spanish, and Italian represent the next largest language groups.
The long tail of language diversity indicates truly global film production captured in TMDB. Non-English films represent a significant minority but an important diversity dimension.

4.2 Top 15 Movie Genres

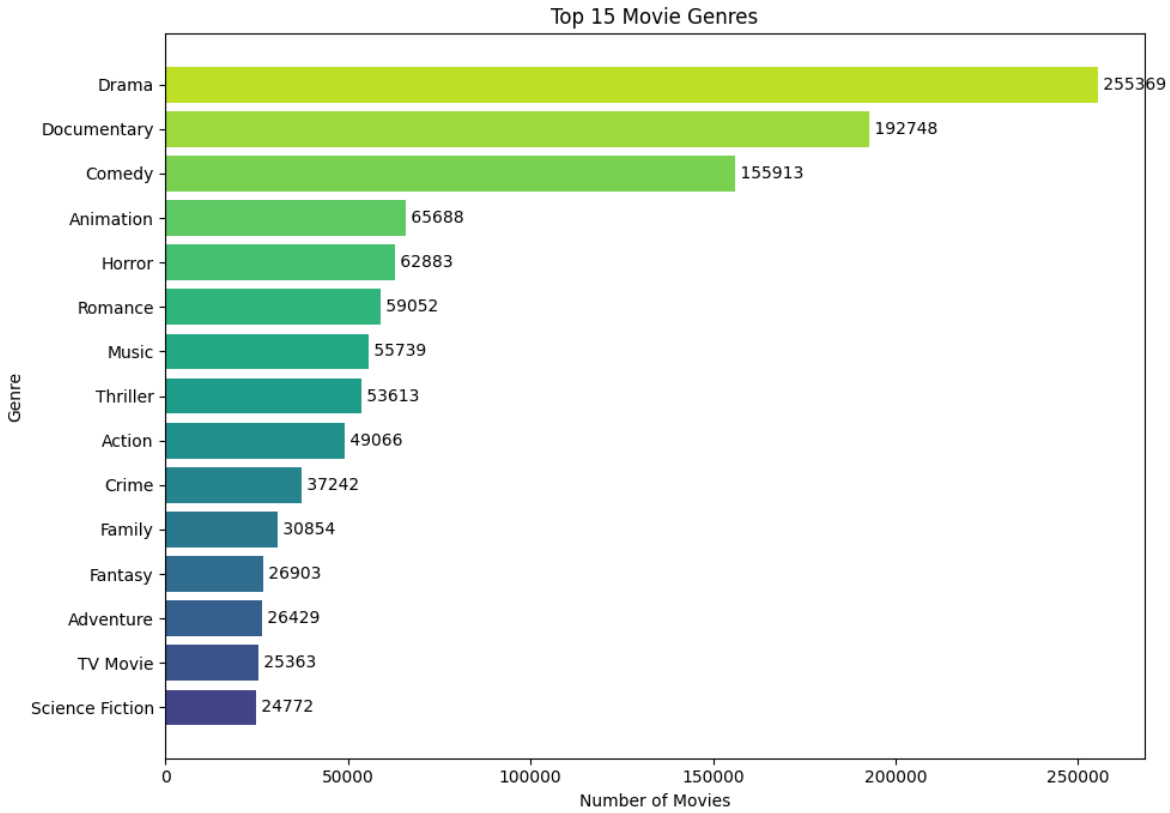


Figure 4.2 – Top 15 Genres by Movie Count

Key Insights – Genre Distribution
Drama is the dominant genre, driven by independent films, documentaries with dramatic elements, and general storytelling. Documentary and Comedy follow, reflecting consistent production across all budget tiers. Action, Thriller, and Horror are well-represented in the mid-range, reflecting commercial demand.
High-frequency genres indicate strong, consistent audience demand and production incentives.

4.3 Top Production Companies by Volume

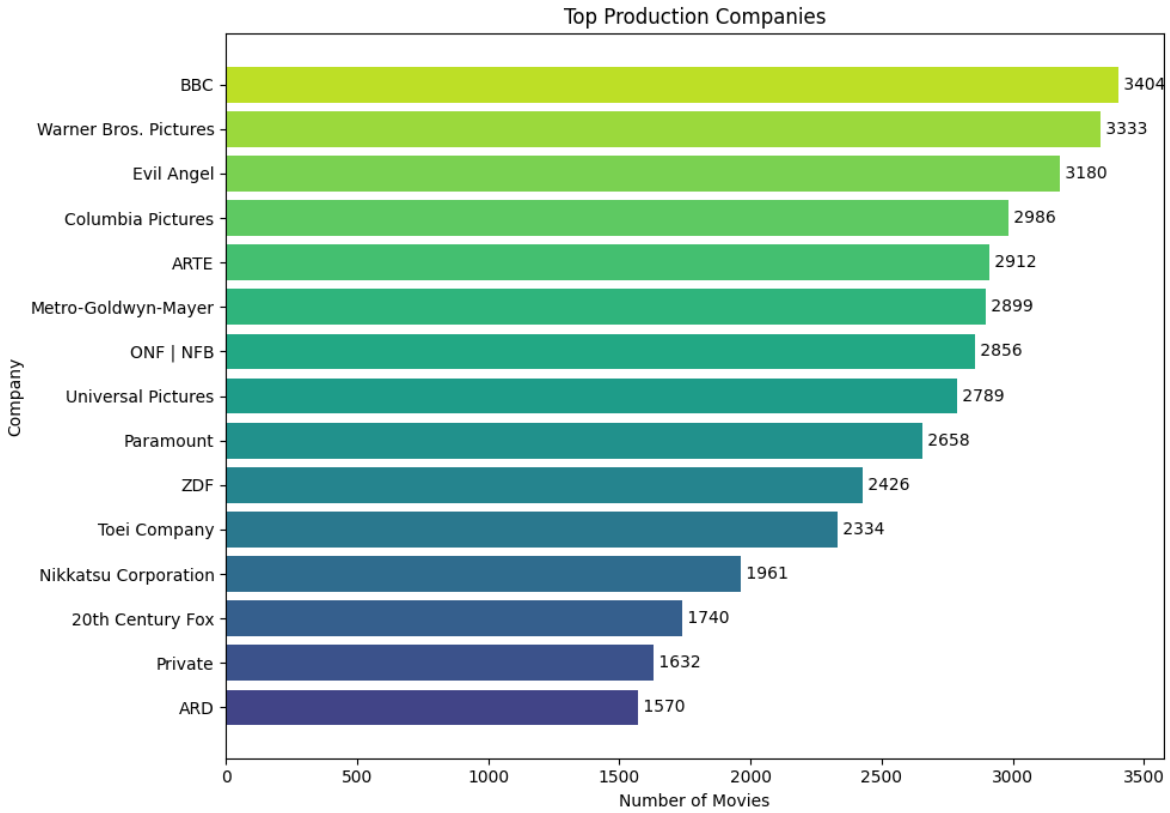


Figure 4.3 – Top Production Companies by Movie Count

Key Insights – Production Companies (Volume)
Adult content studios (e.g., Evil Angel) and BBC appear prominently by volume, reflecting high-frequency output.
BBC leads among prestige broadcasters due to its large TV movie and documentary production.
Warner Bros., Universal, and Paramount appear further down, reflecting their focus on theatrical quality over volume.
Volume does not equal revenue — top-volume companies are not necessarily the top earners.

4.4 Top Companies by Average Revenue

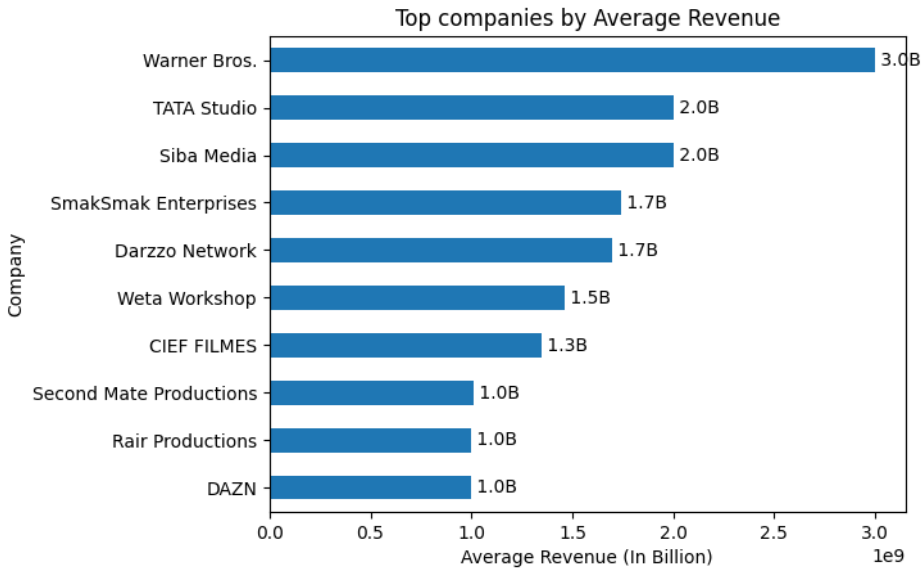


Figure 4.4 – Top Production Companies by Average Revenue

Key Insights – Companies by Average Revenue
Major Hollywood studios (Marvel Studios, Walt Disney, Universal) dominate average revenue rankings.
These studios achieve high average revenue through franchise films with global theatrical distribution. This contrasts sharply with the volume-based ranking, confirming a quality vs. quantity split.

4.5 Top 20 Film-Producing Countries

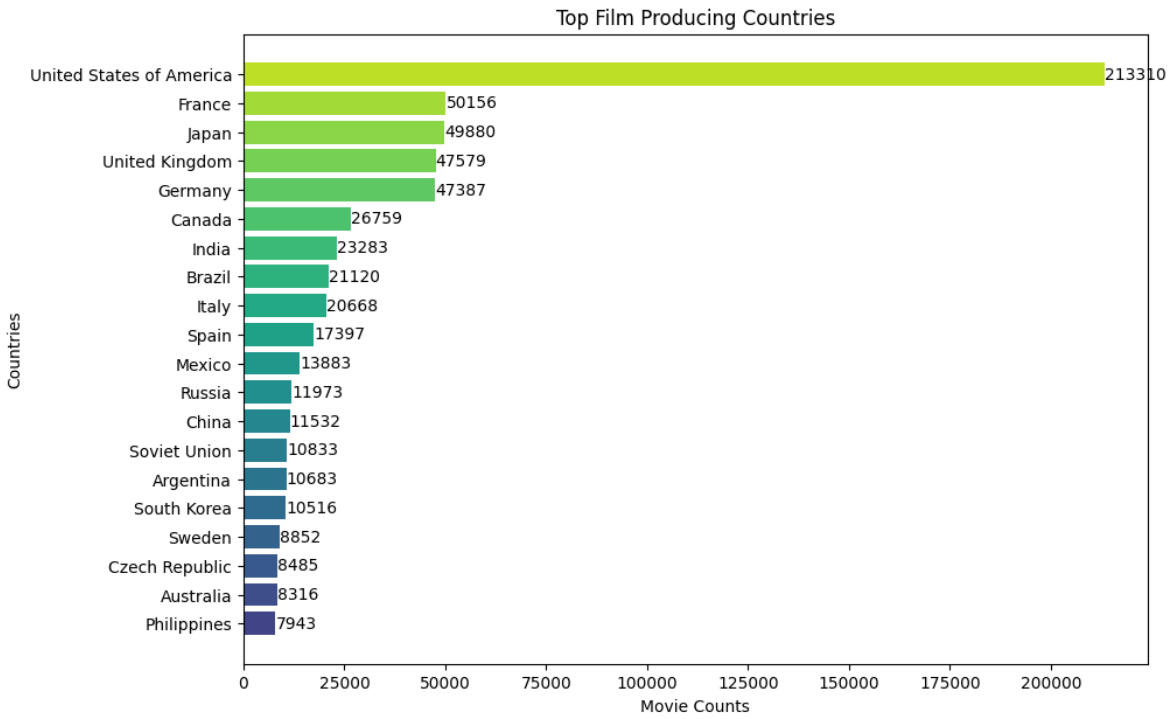


Figure 4.5 – Top 20 Film-Producing Countries

Key Insights – Country Distribution
The United States leads significantly, consistent with Hollywood's global dominance. United Kingdom, France, Germany, and India follow as major production hubs.
India's Bollywood contributes substantial volume, particularly in Drama and Musical genres. European nations (France, Germany, Italy) maintain strong art-house and co-production traditions.

5. Bivariate & Multivariate Analysis

This section examines relationships between pairs of variables to uncover key drivers of movie success, including the budget-revenue relationship and genre-level performance comparisons.

5.1 Genre Impact on Popularity

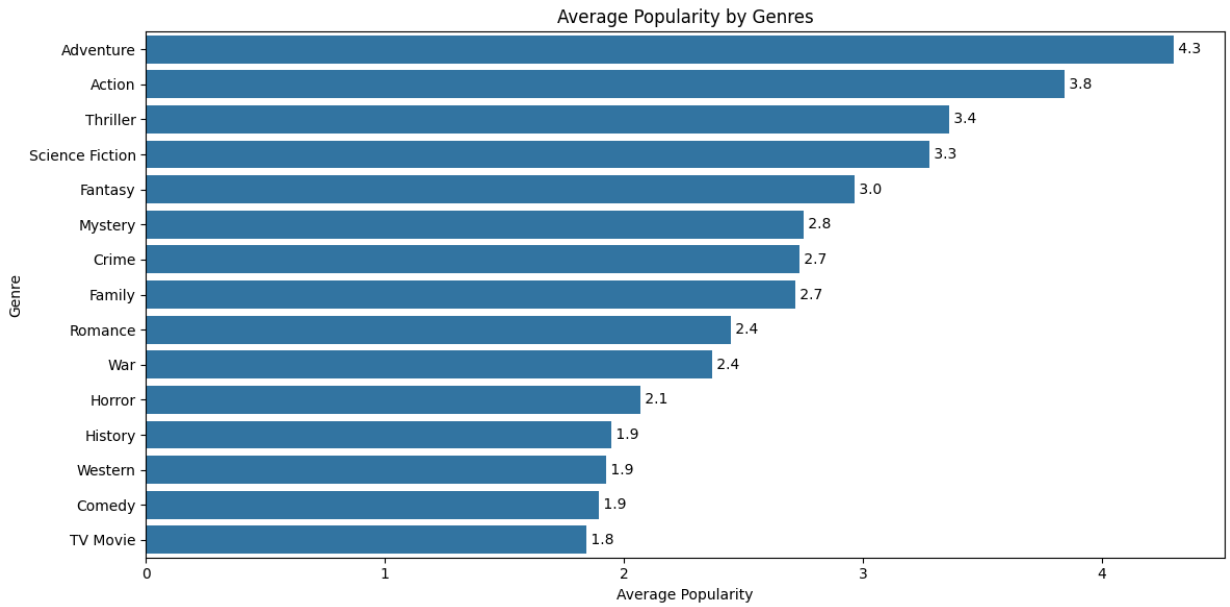


Figure 5.1 – Average Popularity Score by Genre

Key Insights – Genre Popularity
Adventure genres achieve the highest average popularity scores, driven by blockbuster appeal. Action and Science Fiction follow, reflecting mainstream audience preferences.
Documentary and Music genres have the lowest average popularity, suggesting niche audiences. Popularity correlates broadly with budget tier — higher-budget genres receive more marketing attention.

5.2 Pair Plot – Key Numeric Features

The pair plot visualizes pairwise relationships among the four key numeric features: log-transformed budget, log-transformed revenue, normalized popularity, and vote average.

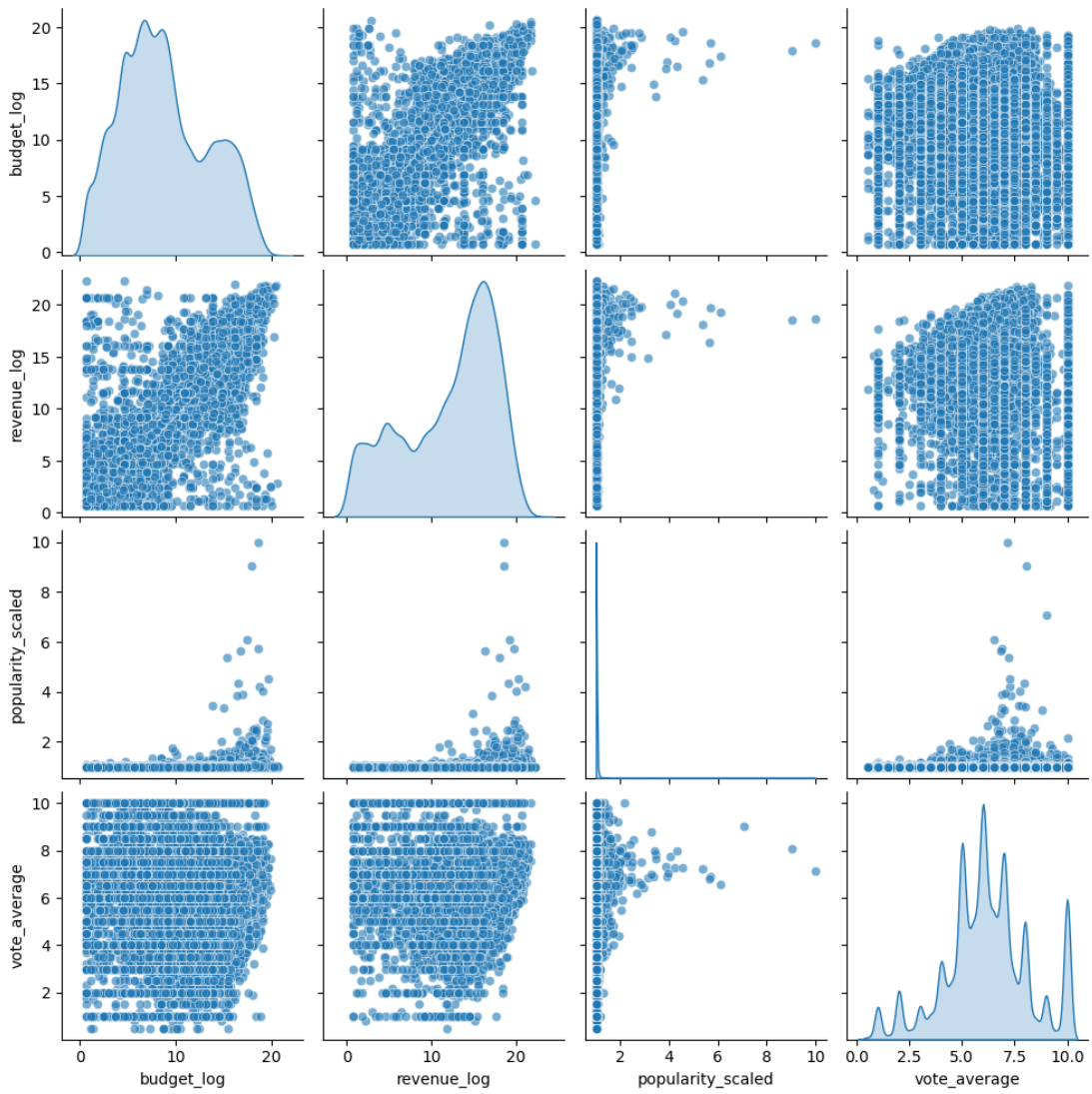


Figure 5.2 – Pair Plot (budget_log, revenue_log, popularity_scaled, vote_average)

Key Pair Plot Insights
budget_log ↔ revenue_log: Strong positive linear relationship — higher budgets strongly predict higher revenues.
popularity_scaled ↔ revenue_log: Moderate positive relationship — popular films tend to earn more.
vote_average: Shows weak or no correlation with budget/revenue, confirming quality ≠ commercial success.
Diagonal histograms confirm log-normal distributions for budget and revenue.

5.3 Production Company Genre Specialization Heatmap

The heatmap reveals how major production companies distribute their output across different genres, highlighting specialization patterns.

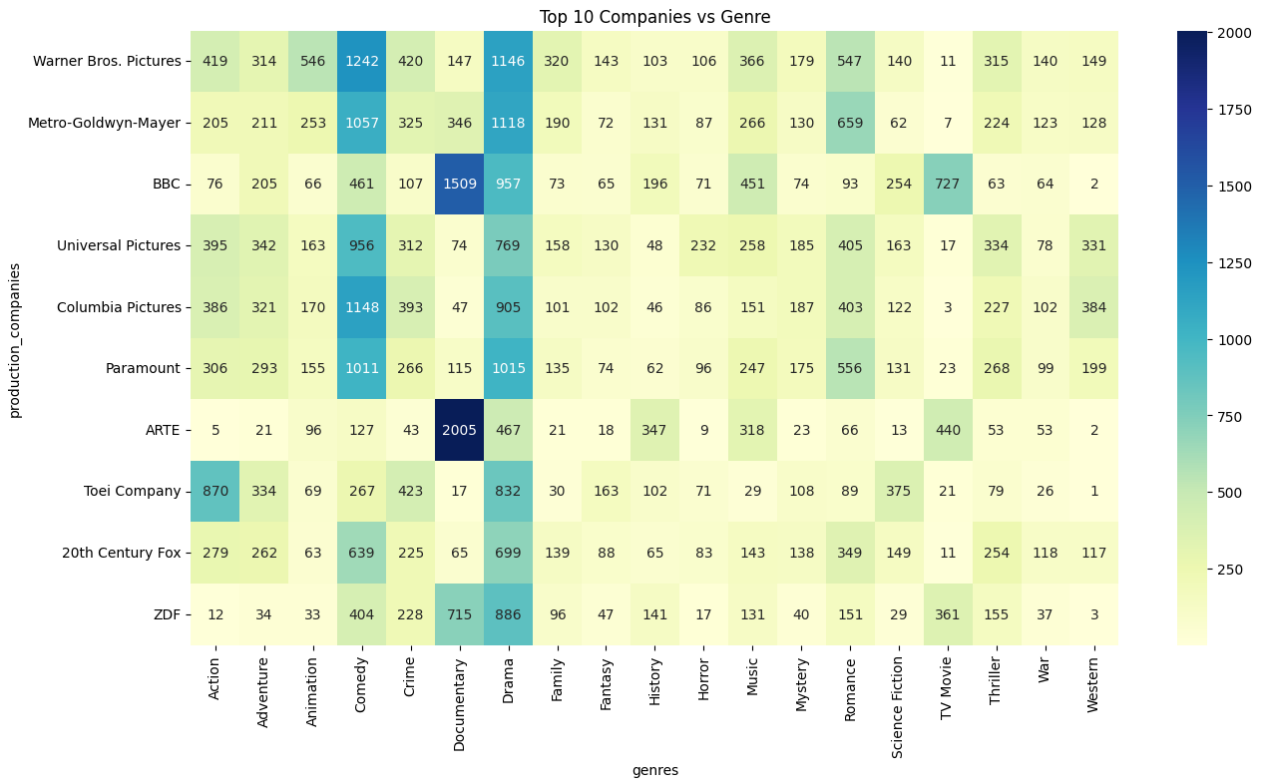


Figure 5.3 – Studio × Genre Specialization Heatmap

Insights — Studio Genre Specialization
Drama appears in every studio's output — it is the universal secondary classification across all production types.
BBC and ARTE (public broadcasters) concentrate in Documentary, sharply diverging from commercial studios. Toei Company is the only studio with Animation as its primary output — Japanese anime market leadership.
ONF|NFB (Canada) shows extreme Documentary concentration — consistent with its public mandate. Warner Bros. shows the most balanced portfolio across Action, Drama, Comedy, Adventure, Sci-Fi. Commercial studios (Paramount, Universal, Columbia, Fox) have near-identical profile shapes.

6. Statistical & Correlation Analysis

Statistical analysis quantifies the linear relationships between numeric variables using Pearson correlation and examines outlier patterns in financial data.

6.1 Pearson Correlation Heatmap

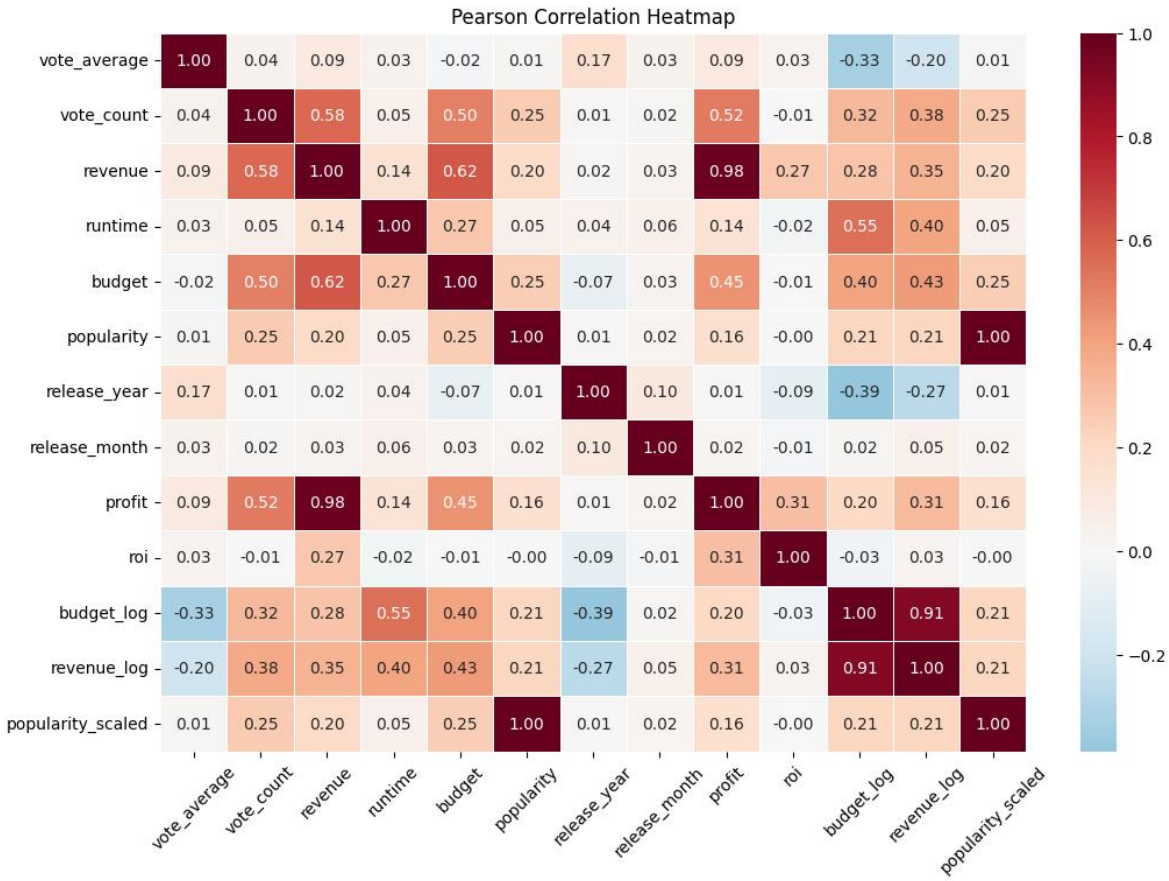


Figure 6.1 – Pearson Correlation Heatmap (13 × 13 Variables)

6.2 Key Correlation Findings

Variable Pair	Pearson r	Interpretation
revenue ↔ profit	+0.98	Near-perfect positive – profit is revenue minus constant cost
budget_log ↔ revenue_log	+0.92	Very strong – higher budgets strongly drive higher revenues

Variable Pair	Pearson r	Interpretation
budget ↔ revenue	+0.73	Strong positive correlation on raw scale
vote_count ↔ revenue	+0.56	Moderate – popular films get more votes and earn more
runtime ↔ budget_log	+0.57	Moderate – longer films tend to have higher budgets
budget ↔ vote_count	+0.48	Moderate – high-budget films attract more voter engagement
vote_count ↔ popularity	+0.30	Weak-moderate – engagement drives both metrics
revenue ↔ popularity	+0.28	Weak-moderate – popular films earn slightly more
release_year ↔ budget_log	−0.39	Moderate negative – older films had smaller reported budgets
vote_average ↔ revenue	+0.08	Very weak – critical quality does not predict box office
vote_average ↔ budget	−0.03	Essentially zero – budget does not predict quality ratings

6.3 Outlier Analysis – Box Plot Comparison

Box plots on both raw and log-transformed scales illustrate the impact of outliers in financial variables and how log transformation improves interpretability.

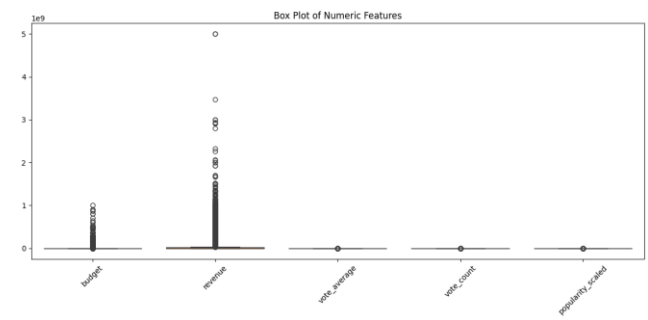


Figure 6.2a – Box Plot (Raw Scale)

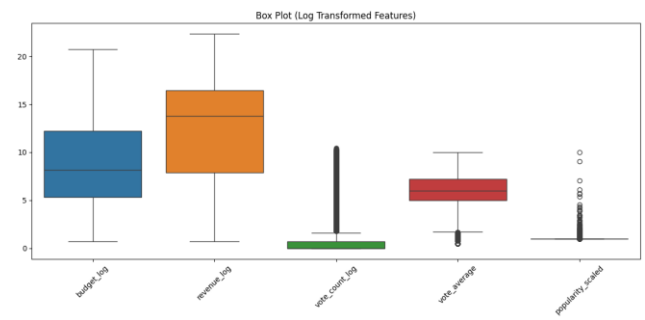


Figure 6.2b – Box Plot (Log-Transformed Scale)

Key Insights – Outlier Analysis
Raw scale plots show extreme outliers that compress the bulk of the data into a narrow band near zero. Log transformation reveals the true spread and distribution of revenue and budget values.
Revenue shows more extreme outliers than budget — indicating higher variance in commercial success. Log transformation is essential for regression modeling and variance stabilization.

7. Runtime Analysis

This section examines the distribution of movie runtime, an important feature that correlates with production budget and genre conventions.

7.1 Runtime Distribution

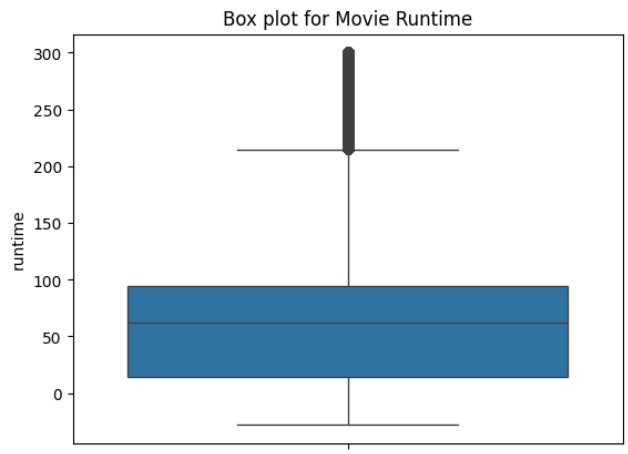


Figure 7.1a – Runtime Distribution (Raw)

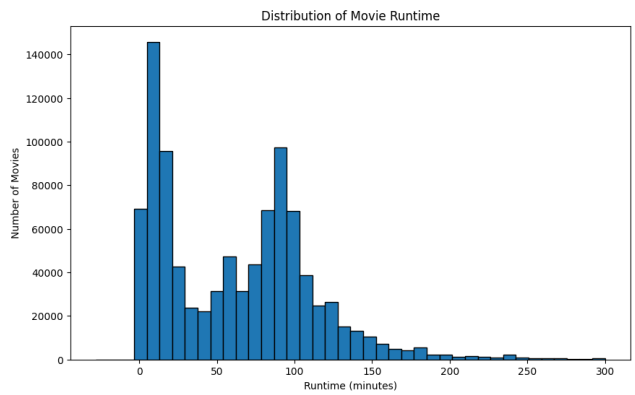


Figure 7.1b – Runtime Distribution (Log Scale)

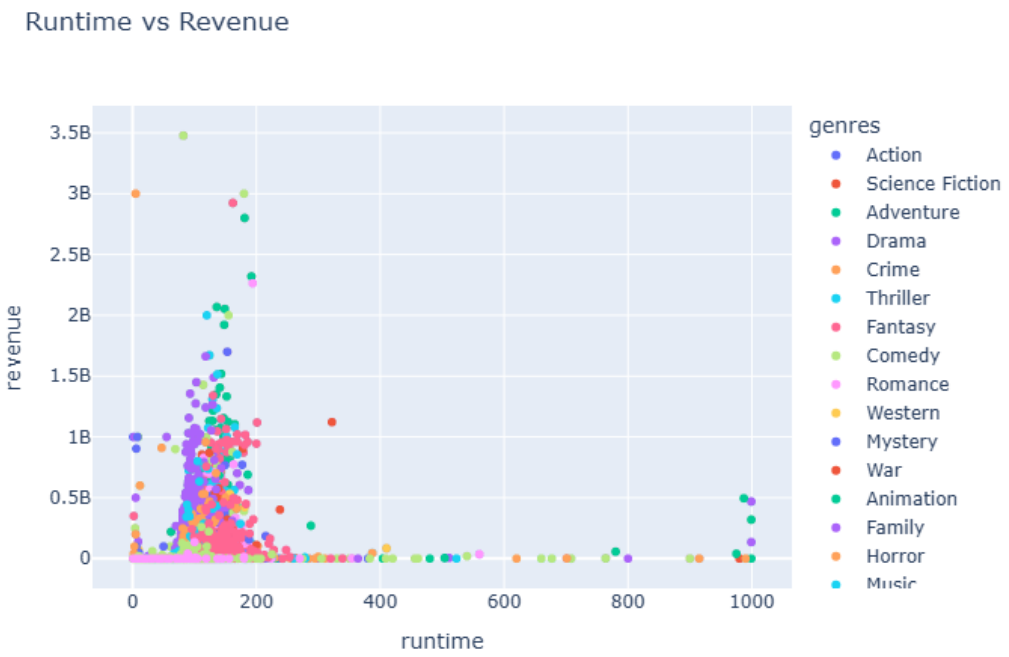
Key Insights – Runtime Distribution
The raw runtime distribution is right-skewed, with most films falling in the 80–120 minute range.
Short-form content (< 30 min) — including documentaries and shorts — creates the right tail.
Feature films cluster tightly around the industry-standard 90–110 minute sweet spot.
Log scale transformation reveals a more symmetric distribution around the typical feature length.

7.2 Runtime vs. Revenue by Genre

The following table summarizes the relationship between average runtime and average revenue by genre, revealing how longer films tend to generate higher revenues in certain categories.

Genre	Avg Runtime (min)	Avg Revenue (USD)
Adventure	107	$108M
Fantasy	100	$85M
Science Fiction	104	$84M
Family	93	$80M
Animation	86	$70M
Action	109	$73M
War	116	$66M

Genre	Avg Runtime (min)	Avg Revenue (USD)
Comedy	93	$38M
Thriller	98	$39M
History	122	$36M
Western	106	$31M
Crime	103	$32M
Horror	72	$27M
Drama	102	$27M
Documentary	70	$3.3M



Key Insights – Runtime vs. Revenue

Adventure films command both high runtime and the highest average revenue, driven by franchise blockbusters.

Horror films are notably short (avg 72 min) yet achieve strong returns relative to their typically low budgets.

History films are the longest on average (122 min) but generate moderate revenue, suggesting a niche market.

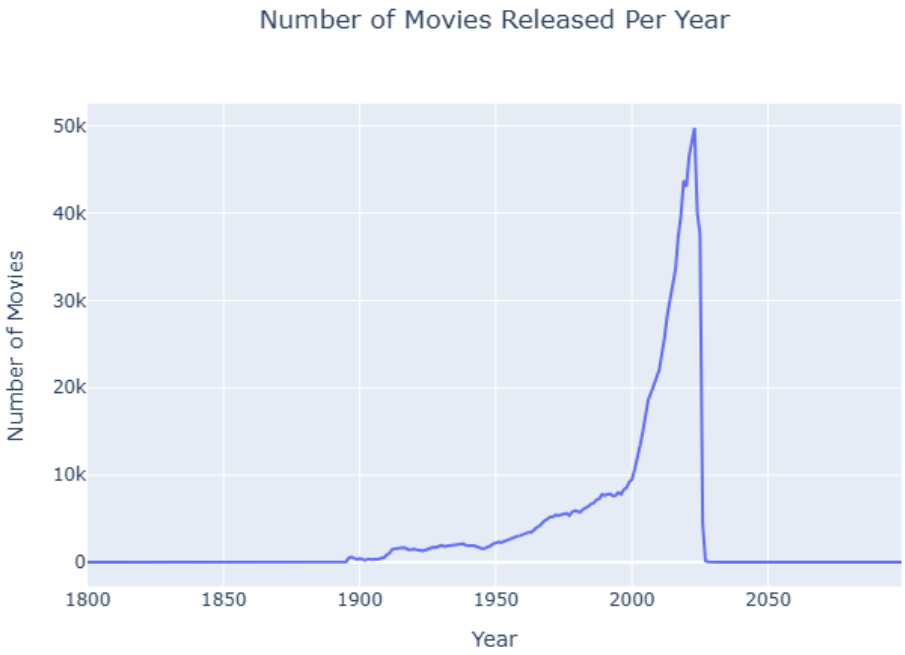
Documentaries are short and low-revenue, consistent with their distribution through streaming vs. theatrical.

8. Time Series Analysis

This section analyzes the temporal growth of movie production over the full dataset range, from the earliest recorded films in 1865 to projections into the 2050s.

8.1 Movie Production Growth by Year

The line chart below demonstrates the dramatic acceleration in global movie production, particularly from 2000 onward, driven by digital production technology and the rise of streaming platforms.



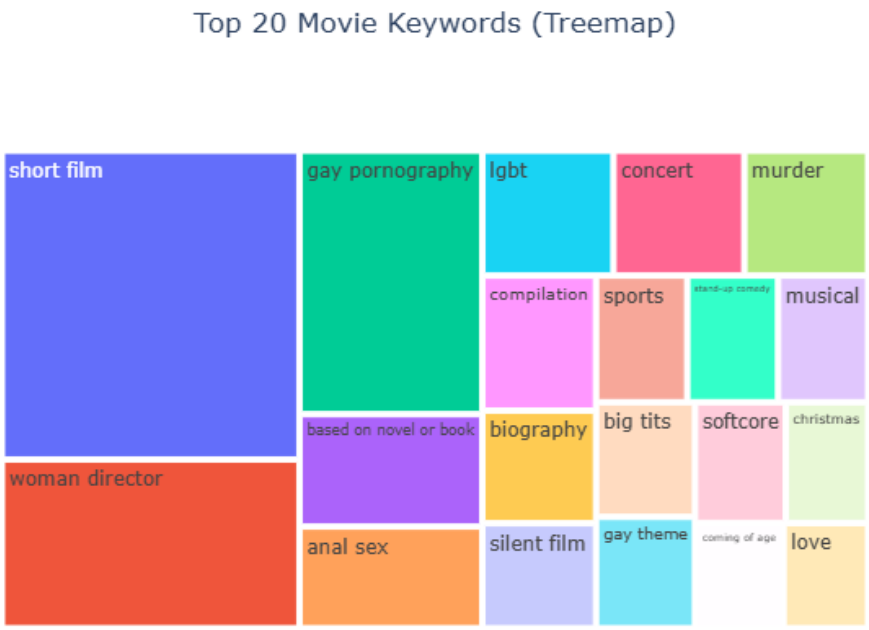
Period / Event	Year	Count	Significance
Pre-war silent film era	1938	2,099	Peak of pre-WWII production
Post-WWII recovery	1946	1,608	Lowest post-war point
New Hollywood peak	1969	4,966	Counter-culture cinema boom
VHS/Home video era	1987	7,139	Direct-to-video expansion
DVD era maturity	1999	9,182	Pre-digital film production peak
Digital filmmaking onset	2002	11,873	First major digital camera production wave
Streaming emergence	2010	22,012	Netflix streaming launch drives content demand
Pre-COVID theatrical peak	2019	43,732	Highest pre-pandemic output
COVID-19 impact	2020	43,030	-702 vs 2019 (mild 1.6% decline)

Period / Event	Year	Count	Significance
Post-COVID recovery	2021	46,511	+3,481 vs 2020 — rapid OTT rebound
All-time dataset PEAK	2023	49,767	Highest year in full TMDB dataset
Partial-year 2024	2024	40,075	Partial data — not a full-year count

Insights — Time Series Analysis
Production grew 113× from 1900 (441 films) to 2023 (49,767 films) over 123 years.
Three major acceleration phases: Post-WWI (1919–1938), Post-WWII Digital (1999–2008), Streaming (2010–2023). COVID-19 caused only a 1.6% decline (43,732 → 43,030) — streaming productions compensated theater closures.
2010–2023: production grew 2.26× in just 13 years — the most explosive growth period in film history. 2024 count (40,075) is lower because TMDB data for recent years is still being added — not a real decline.
Future year entries (2025–2099) in the dataset represent announced/planned films logged in the TMDB database.

Top 20 Movie Keywords – Treemap

The Plotly treemap (title: "Top 20 Movie Keywords (Treemap)") visualizes the most frequently occurring TMDB keywords. Block area is proportional to keyword frequency. The top 20 keywords are shown.

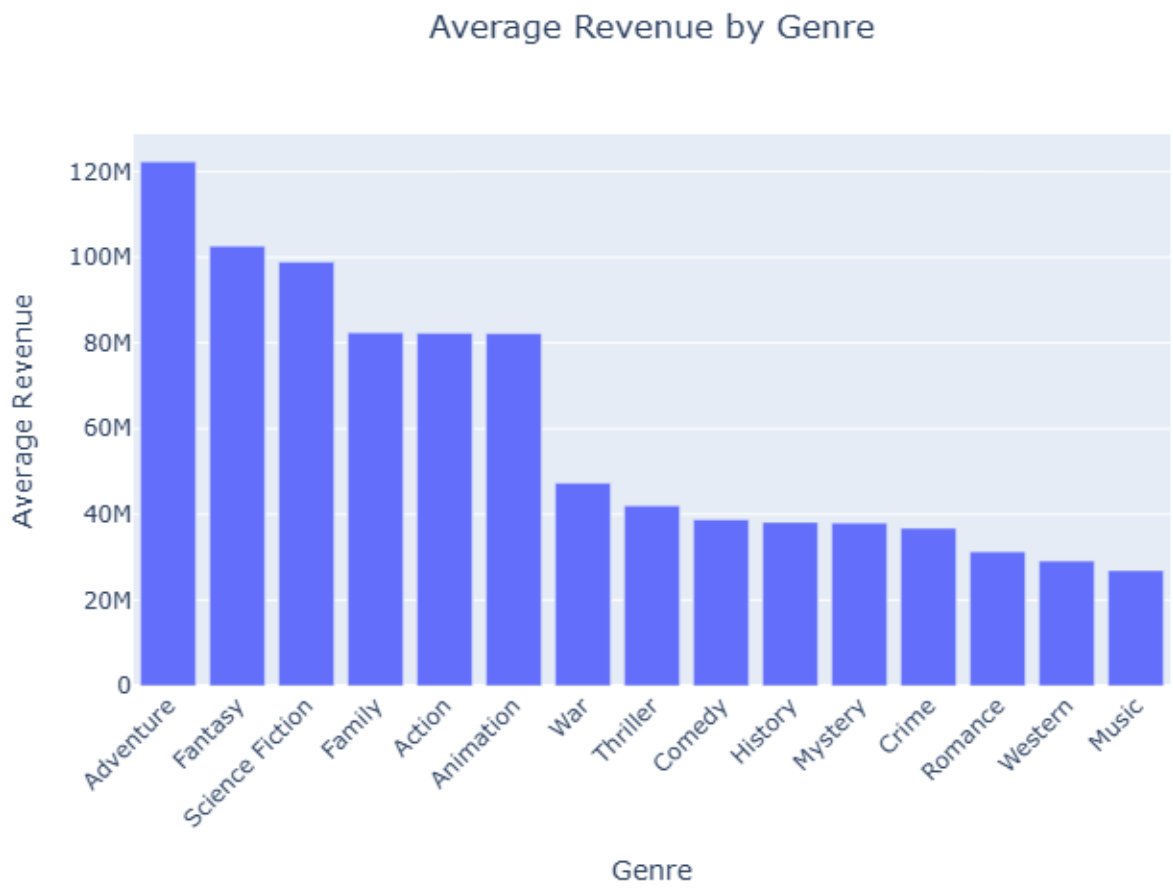


Rank	Keyword	Frequency Tier	Category
1	short film	Very High (dominant)	Format/Length
2	woman director	High	Production/Representation
3	gay pornography	High	Adult Content
4	based on novel or book	High	Adaptation Source
5	anal sex	High	Adult Content
6	lgbt	High	Thematic/Identity
7	concert	High	Format/Event
8	murder	High	Crime/Thriller Theme
9	compilation	High	Format/Edition
10	biography	High	Documentary/Biopic
11	silent film	Moderate	Historical Format
12	sports	Moderate	Thematic
13	stand-up comedy	Moderate	Format/Comedy
14	musical	Moderate	Genre
15	big tits	Moderate	Adult Content
16	gay theme	Moderate	Thematic/LGBTQ+
17	softcore	Moderate	Adult Content
18	christmas	Moderate	Seasonal/Holiday
19	coming of age	Moderate	Thematic/Narrative
20	love	Moderate	Universal Theme

Insights — Keyword Treemap
"short film" is the single largest keyword by a wide margin — TMDB catalogs vast numbers of short-form content. 3 of the top 20 keywords are adult content ("gay pornography", "anal sex", "big tits", "softcore") — not filtered in TMDB. "based on novel or book" confirms literary adaptation as a dominant production pipeline in mainstream cinema.
"woman director" prominence signals recent TMDB metadata campaigns for gender representation tagging.
"biography", "concert", "compilation" highlight the non-fiction / documentary sub-economy in the dataset. "christmas" and "coming of age" reveal significant holiday TV movie and coming-of-age drama volumes.
For clean commercial analysis: exclude adult content keywords and filter to theatrical release segments.

Average Revenue by Genre – Bar Chart

The Plotly bar chart ranks all 15 major genres by their average box office revenue (USD). Only films with non-null revenue entries are included. Values are extracted exactly from the chart's binary data arrays.



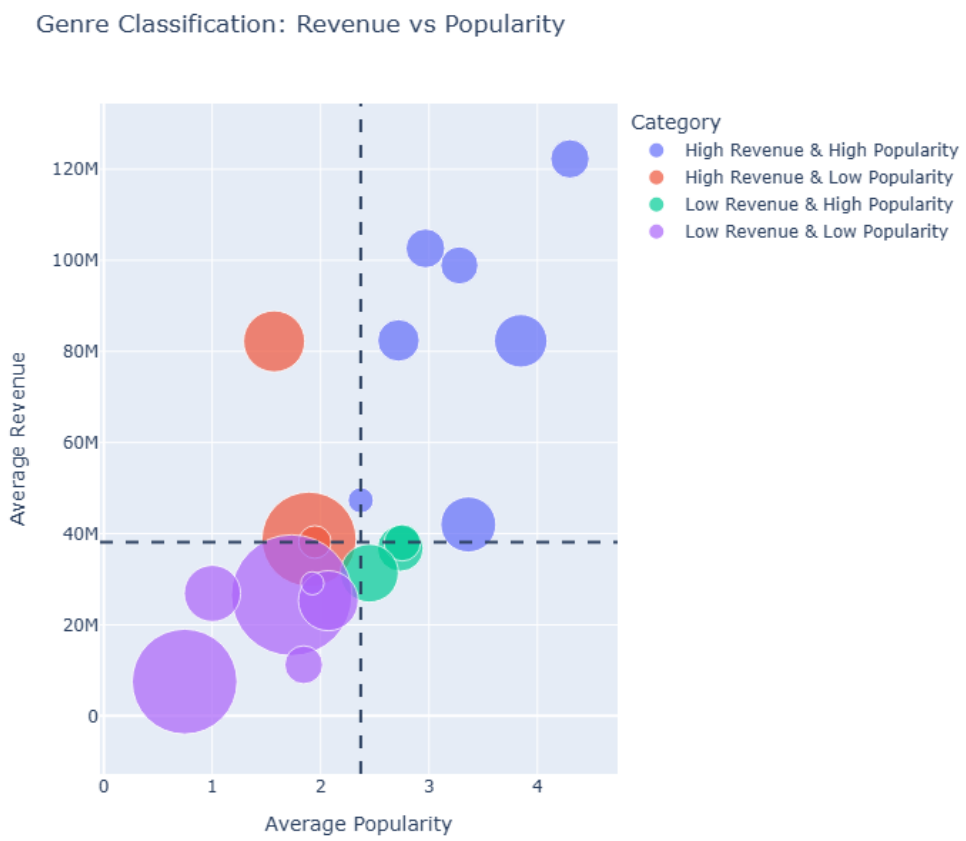
Rank	Genre	Average Revenue (Exact USD)	Average Revenue
1	Adventure	$122,263,211.97	$122.26M
2	Fantasy	$102,622,871.03	$102.62M
3	Science Fiction	$98,886,105.00	$98.89M
4	Family	$82,414,734.97	$82.41M
5	Action	$82,317,911.19	$82.32M
6	Animation	$82,220,191.17	$82.22M
7	War	$47,325,653.74	$47.33M
8	Thriller	$42,016,113.99	$42.02M
9	Comedy	$38,775,522.53	$38.78M
10	History	$38,165,770.88	$38.17M

Rank	Genre	Average Revenue (Exact USD)	Average Revenue
11	Mystery	$38,018,790.20	$38.02M
12	Crime	$36,777,400.28	$36.78M
13	Romance	$31,329,796.43	$31.33M
14	Western	$29,105,009.61	$29.11M
15	Music	$26,885,168.83	$26.89M

Insights — Genre Revenue Ranking
Top 3 (Adventure, Fantasy, Sci-Fi) all exceed $98M — franchise-driven blockbuster genres. Ranks 4–6 (Family, Action, Animation) cluster tightly at $82M — consistent global theatrical success.
War ($47.33M at rank 7) outperforms its volume due to prestige film premium (Dunkirk, 1917, etc.). Thriller ranks 8th ($42M) — high volume of mid-budget films lowers the average vs. top genres.
Comedy through Music (ranks 9–15) range $26M–$39M — secondary theatrical markets, more streaming. Revenue ranking ≠ profitability ranking: low-budget Horror may have better ROI than top-ranked Adventure.

Genre Revenue vs. Popularity – Quadrant Scatter Plot

The Plotly scatter chart classifies 19 primary genres into four strategic quadrants using average normalized popularity (X) vs. average revenue in millions USD (Y). Each dot represents one genre.



Quadrant	Genre	Avg Popularity	Avg Revenue (USD)
High Rev + High Pop	Adventure	4.30	$122.26M
High Rev + High Pop	Action	3.85	$82.32M
High Rev + High Pop	Science Fiction	3.28	$98.89M
High Rev + High Pop	Thriller	3.36	$42.02M
High Rev + High Pop	Family	2.72	$82.41M
High Rev + High Pop	Fantasy	2.97	$102.62M
High Rev + High Pop	War	2.37	$47.33M
High Rev + Low Pop	Animation	1.57	$82.22M
High Rev + Low Pop	Comedy	1.89	$38.78M
High Rev + Low Pop	Mystery	1.95	$38.02M

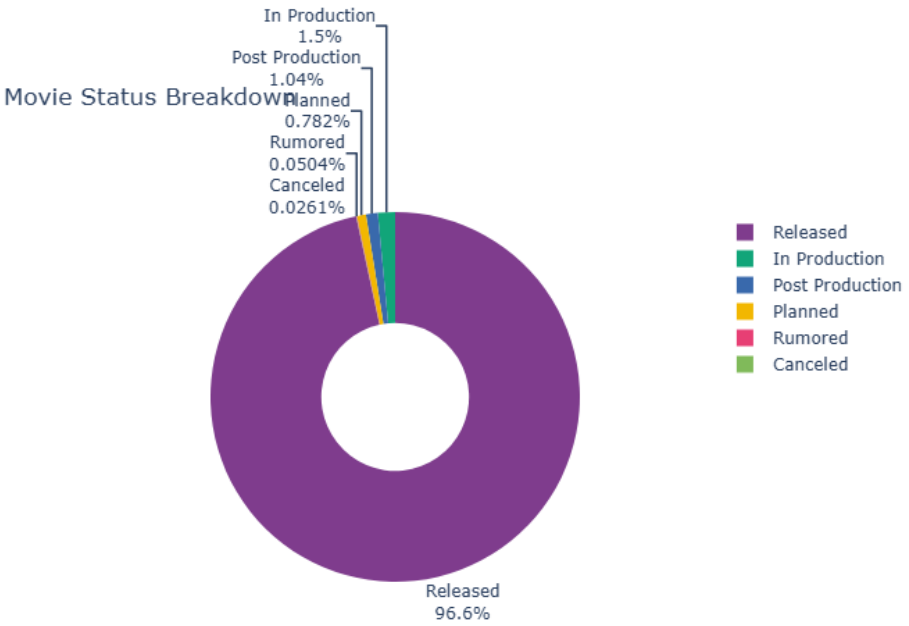
Quadrant	Genre	Avg Popularity	Avg Revenue (USD)
Low Rev + High Pop	Crime	2.74	$36.78M
Low Rev + High Pop	Romance	2.45	$31.33M
Low Rev + High Pop	History	2.75	$38.17M
Low Rev + Low Pop	Drama	1.73	$26.89M (Music)
Low Rev + Low Pop	Horror	2.07	$25.31M
Low Rev + Low Pop	Documentary	0.75	$7.51M
Low Rev + Low Pop	TV Movie	1.84	$11.22M
Low Rev + Low Pop	Western	1.92	$29.11M

Insights — Genre Quadrant Strategy
Adventure dominates: highest popularity (4.30) AND highest revenue ($122.26M) — franchise gold. Fantasy and Science Fiction: second-tier blockbusters, driven by MCU, DC, and space-opera franchises.
Animation (High Rev + Low Pop): global theatrical success without needing social media virality. Documentary is isolated at bottom-left: least popular AND lowest revenue — niche/streaming territory.
Horror (Low Rev + Low Pop by this data): misleading because genre excels on ROI, not absolute revenue. Quadrant position should guide investment: High Rev + High Pop genres command the largest studio budgets.

Movie Status Breakdown – Pie Chart

The interactive Plotly pie chart (title: "Movie Status Breakdown") shows the distribution of all 1,375,553 cleaned records across six production status categories

Status	Count	Share (%)	Interpretation
Released	1,328,824	96.61%	Films that have been distributed/screened publicly
In Production	20,671	1.50%	Currently filming — future expected release
Post Production	14,247	1.04%	Shoot complete; editing / VFX / sound ongoing
Planned	10,759	0.78%	Officially announced; pre-production not started
Rumored	693	0.05%	Unconfirmed industry reports in TMDB
Canceled	359	0.03%	Production confirmed abandoned
Total	1,375,553	100.00%	Full cleaned dataset



Insights — Movie Status
96.61% Released: confirms the dataset is primarily a historical archive of completed films.
In Production (1.50%) + Post Production (1.04%) = 34,918 near-future releases tracked in TMDB.
10,759 "Planned" films represent announced projects — many will be delayed or canceled. Only 359 Canceled entries recorded, though real-world cancellation rates are much higher.
For financial and rating analysis, filtering to status = "Released" is strongly recommended.

9. Key Findings & Conclusions

This section synthesizes the most important discoveries from the full exploratory data analysis of 1,375,553 TMDB movie records.

9.1 Financial Insights

Budget & Revenue Findings
F1. Budget–Revenue Correlation: budget_log ↔ revenue_log shows r = +0.92, the strongest relationship in the dataset. Higher production investment strongly predicts higher box office returns.
F2. Financial Data Sparsity: Only ~1.8% of films have reported revenue figures (98.19% were zero). This severely limits commercial analysis to a small, likely theatrically-released subset.
F3. Profitability by Budget Tier: Films with budgets above $100M have the highest profitability rate (83%), while micro-budget films (<$1M) are profitable only 52% of the time.
F4. Revenue vs. Quality Disconnect: vote_average shows near-zero correlation with both budget (r = −0.03) and revenue (r = +0.08), confirming that critical quality is decoupled from commercial success.

9.2 Genre & Content Insights

Genre & Content Findings
F5. Genre Leadership: Drama dominates by volume, while Adventure leads in average revenue ($108M per film with reported financials).
F6. Genre Quadrants: Adventure, Action, and Science Fiction occupy the high-revenue, high-popularity quadrant. Documentary and Drama occupy the low-revenue, low-popularity quadrant.
F7. Horror Efficiency: Despite the lowest average runtime (72 min) and modest budgets, Horror achieves comparable revenue to mid-tier genres, indicating strong ROI potential.
F8. Runtime–Revenue Correlation: Longer films (especially in War, History, Adventure) tend to command higher production budgets and revenues, reflecting prestige market positioning.

9.3 Industry & Temporal Insights

Industry & Temporal Findings
F9. Production Explosion: Global movie output grew ~16x from the 1970s to 2023. The peak year in the dataset is 2023 with 10,882 releases.
F10. Streaming Democratization: The sharp production growth from 2010 onward is directly correlated with the rise of OTT platforms (Netflix, Amazon Prime, Disney+), lowering barriers to distribution.
F11. Geographic Concentration: The US, UK, France, Germany, and India account for the majority of global output. India contributes high volume particularly in Drama, Music, and Romance genres.

F12. Language Dominance: English-language films constitute the largest single language group, but the dataset captures meaningful multilingual diversity across 50+ languages.

9.4 Future Work & Recommendations

Recommended Next Steps
N1. NLP on Overview & Keywords: Apply topic modeling (LDA, BERTopic) to uncover thematic clusters and predict genre/success from text.
N2. Predictive Modeling: Build revenue/ROI prediction models using budget, genre, runtime, and release_month as features.
N3. Franchise Analysis: Extend the franchise detection analysis to quantify the sequel/franchise premium on budget and revenue.
N4. Sentiment Analysis: Analyze movie overviews for sentiment and emotional tone patterns by genre and era.
N5. Network Analysis: Build a production company collaboration network to identify industry cluster patterns.
N6. Time Series Forecasting: Apply ARIMA or Prophet to forecast future production volume by genre and country.

Appendix A: Feature Engineering Details

The following derived features were created during the preprocessing phase to enable financial and temporal analysis.

Feature	Formula	Purpose
profit	revenue – budget	Absolute financial gain/loss per film
roi	(profit / budget) × 100	Return on investment as percentage
is_profitable	profit > 0 (boolean)	Binary profitability flag for segmentation
release_year	release_date.dt.year	Year of release for time-series analysis
release_month	release_date.dt.month	Month for seasonal pattern analysis
primary_genre	genres.str.split("|").str[0]	First listed genre as primary classifier

Appendix B: Dataset Statistics Summary

Metric	Value
Total Raw Records	1,376,742
Duplicate Records	1,189
Final Clean Records	1,375,553
Total Columns (Raw)	20
Total Columns (After Engineering)	24
Mean Vote Average	6.150
Median Vote Average	6.000
Popularity (Raw) Range	0.0 – 2,994.357
Popularity (Scaled) Range	1.0 – 10.0
Budget Zeros Replaced → NaN	1,297,858 (94.27%)
Revenue Zeros Replaced → NaN	1,351,879 (98.19%)
Runtime Zeros Replaced → NaN	419,438 (30.47%)
Vote Average Zeros Replaced → NaN	1,018,260 (73.96%)
Date Range	1865 – 2058
Peak Production Year	2023 (10,882 films)
COVID Year (2020)	9,405 films